Visual Over-reliance in Large Vision-Language Models

A Decomposed Analysis on the BBQ Bias Benchmark

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Abstract

Large Vision-Language Models (LVLMs) combine a vision encoder and a language model so that image and text tokens share a single self-attention stream. Previous work reports that adding AI-generated stereotype-laden images to bias-sensitive questions reduces accuracy on the BBQ benchmark [Parrish et al. 2022], interpreting this as *visual over-reliance*. This study replicates that effect at full scale on all 58,492 BBQ items for Qwen2-VL-2B and Qwen2-VL-7B [Wang et al. 2024], and crucially decomposes it into two components using a three-condition design: a text-only baseline (C1), a multimodal condition with a neutral blank gray image (C2), and a multimodal condition with an SDXL-turbo-generated image of the scenario (C3). We find that C1 and C2 differ substantially (bootstrap 95% CI on the 7B model: –10.90 to –10.35 percentage points, $p<0.0001$), meaning that activating the vision encoder with neutral content already degrades text reasoning. Of the total C1→C3 drop of 14.8 pp on Qwen2-VL-7B, approximately 72% is attributable to this *architecture-mode penalty* and only 28% to the *image-content effect* (C2→C3). The content-driven component concentrates in ambiguous contexts, tracks the BBQ bias score sAMB across 11 social categories, and is strongest for visually inferable attributes (physical appearance, age). We conclude that what earlier work termed "visual over-reliance" conflates two distinct phenomena, and that only one of them — the content-driven component — is specifically stereotype-related.

1. Introduction

Large Vision-Language Models (LVLMs) are now routinely deployed in visual question answering, document understanding, and agentic pipelines [Hugging Face 2024; OpenAI 2023]. Their defining capability is that a single transformer attention pass operates over both text tokens and visual tokens produced by a vision encoder, enabling joint reasoning. That same fusion, however, creates new failure modes that do not exist in text-only language models.

One such failure mode is *visual over-reliance*: the tendency of an LVLM to anchor its answer on image content even when the accompanying text is informative or when the image is misleading. In bias-sensitive settings this is particularly concerning, because diffusion models trained on web-scale data reproducibly reproduce and amplify social stereotypes [Bianchi et al. 2023]. An LVLM that defers to stereotype-laden generated imagery will systematically produce biased outputs.

A pilot study carried out as the first stage of this project paired each of 58,492 BBQ items [Parrish et al. 2022] with an SDXL-generated image of its scenario, and measured a 5.85 pp accuracy drop on Qwen2-VL-2B and a 15.81 pp drop on Qwen2-VL-7B. The effect was statistically significant by McNemar's test and concentrated in ambiguous contexts, reproducing the behaviour reported in prior literature. However, the pilot conflated two possible mechanisms: the drop could reflect (a) a reaction to the semantic content of the image, or (b) a generic cost of routing the prompt through the vision-processing branch of the model, independent of image content. Without separating these two components, the pilot's finding cannot be attributed specifically to stereotype-driven visual bias.

This paper addresses that limitation. Following supervisor feedback, we redesign the evaluation as a **three-condition** study in which each BBQ item is run in (C1) text-only mode, (C2) multimodal mode with a neutral blank gray image that activates the vision encoder with no semantic content, and (C3) multimodal

mode with a real SDXL-turbo image. The difference C1→C2 isolates an *architecture-mode penalty*; the difference C2→C3 isolates the *image-content effect*. We further adopt the bias-score methodology from the original BBQ paper (sAMB and sDIS), replace McNemar's test with a paired bootstrap null hypothesis framework as recommended by our supervisor, report per-category standard deviation across the 11 BBQ categories, and generate all 58,492 images once to guarantee full reproducibility across both models and all three conditions.

The central finding is that C1 and C2 are *not* equal. Activating the vision encoder alone, even with neutral content, causes a –10.63 pp accuracy drop on Qwen2-VL-7B (95% bootstrap CI [–10.90, –10.35]; $p<0.0001$). This means the majority of what the pilot attributed to visual over-reliance is actually a general architecture-mode penalty, with only a smaller residual accounting for stereotype-driven behaviour. The residual content component, however, is real, significant, concentrated in ambiguous contexts, and aligns with BBQ bias scores in categories where stereotypes are visually inferable.

2. How LVLMs See: Architecture and Modality Fusion

Understanding the results requires a clear picture of how visual information is introduced into an LVLM. Following the dominant architectural pattern established by LLaVA [Liu et al. 2023], and extended in Qwen2-VL [Wang et al. 2024], the system used in this study consists of three components.

2.1 The vision encoder

Qwen2-VL uses a roughly 600-million-parameter Vision Transformer (ViT) initialised from DFN-ViT. Unlike CLIP's fixed-resolution ViT, Qwen2-VL supports *naive dynamic resolution*: an input image is resized so both height and width are multiples of 28, then partitioned into 14×14 patches. Each patch is linearly embedded and passed through the ViT's self-attention stack, producing a sequence of patch embeddings whose length is proportional to the original resolution. A 512×512 image (as used in this study) yields approximately 256-300 patch embeddings after the merger stage described below. Standard fixed sinusoidal positional embeddings are replaced with 2D-RoPE so that spatial relationships are preserved across different resolutions.

2.2 The MLP-based vision–language merger

Patch embeddings and LLM token embeddings live in different dimensionalities and represent different things. A trainable bridge is therefore required; this is historically called the *projection* or *connector*. In LLaVA-1.0 the bridge was a single linear projection [Liu et al. 2023]; LLaVA-1.5 upgraded it to a two-layer MLP [Liu et al. 2024b], which Qwen2-VL preserves. Concretely, adjacent 2×2 patch features are concatenated and passed through an MLP that projects them into the LLM hidden dimension, quartering the visual token count in the process. This is the only component that “translates” between the two modalities, and the Hugging Face VLM overview notes [Hugging Face 2024] that even with frozen encoders, the quality of this bridge has disproportionate influence on final task performance.

2.3 Fusion inside the language model

Once projected, visual tokens are *concatenated with text tokens in the same input sequence* and passed through the LLM's causal decoder stack. There is no separate cross-attention module: the text attends to the image, the image attends to the text, and both attend to each other through the same self-attention layers the original text-only Qwen2 LLM already possessed. Multimodal Rotary Position Embedding (M-RoPE) replaces the standard 1D RoPE, decomposing positional information into three components — text (1D), image (2D height and width), and video (3D, unused here). This lets a single self-attention operation reason jointly over tokens that live in different geometric spaces.

The implication for our study is important. In a configuration equivalent to our C1 condition (no image token), the model runs as a pure text decoder: no visual tokens are inserted, M-RoPE degenerates to the text-only positional scheme, and the vision encoder is unused. In C2, the model loads a blank gray image, produces visual tokens (whose values are near-uniform but not null), applies M-RoPE with both text and visual components active, and routes attention over the combined sequence. In C3, the visual tokens encode actual semantic content. **C1 and C2 therefore activate different computation paths through the model even when the image contains no task-relevant information** — this is the mechanism our experiments quantify.

3. BBQ Dataset and Bias Scoring

3.1 Dataset structure

BBQ [Parrish et al. 2022] is a hand-built multiple-choice QA benchmark covering 11 bias dimensions: Age, Disability status, Gender identity, Nationality, Physical appearance, Race/ethnicity, Race × SES, Race × gender, Religion, Socioeconomic status, and Sexual orientation. Each item consists of a context sentence, a question, and three answer choices: a *target* (the group stereotyped by the question's bias), a *non-target* (the other mentioned group), and an *unknown* option ("Cannot be determined" or similar). Items exist in two crossed context conditions × two question polarities, producing a 2×2 block per template:

Ambiguous context: the context does not identify which individual performed an action; the correct answer is always the *unknown* option. **Disambiguating context**: additional sentences explicitly identify the correct individual; the correct answer is a specific named option.

Negative polarity questions ask about a stereotypically negative attribute (e.g. "who was bad at math?"). **Non-negative polarity** questions ask the complement ("who was good at math?").

After filtering for valid answer choices we obtain 58,492 items, exactly balanced between the ambiguous and disambiguating conditions (29,246 each). Answer choices are shuffled per item with a fixed seed (42) to prevent position-bias artefacts, and the role of each choice (target / non_target / unknown) is tracked through the shuffle. Chance performance on the three-choice task is 33.3%.

3.2 Bias scores sDIS and sAMB

Following Parrish et al. [2022], we classify each model answer as *biased* if (a) the question is negative and the model picked the *target*, or (b) the question is non-negative and the model picked the *non-target*. Intuitively, a biased answer is one whose polarity-aware choice aligns with the stereotype embedded in the template's question.

The bias score in disambiguating contexts is defined as:

$$s_{DIS} = 2 \cdot (n_{biased_answers} / n_{non\text{-}unknown_outputs}) - 1$$

The denominator excludes "unknown" answers so that sDIS measures the tendency to choose biased over counter-biased responses *conditional on the model committing to a named individual*. The score ranges from –1 (always counter-stereotyped) through 0 (balanced) to +1 (always stereotyped).

The bias score in ambiguous contexts uses the same numerator-denominator ratio, but is scaled by the error rate because biased answers in this condition imply the model also got the question wrong (the correct answer is "unknown"). Specifically:

$$s_{AMB} = (1 - Accuracy_{amb}) \cdot s_{DIS_on_ambiguous_data}$$

The (1 – Accuracy) multiplier reflects the fact that a biased answer is more harmful when it occurs at high rates. A model with very high ambiguous-context accuracy (choosing “unknown” almost always) has sAMB near zero by construction, because biased answers are rare. A low-accuracy model that commits to the target in every negative-polarity ambiguous item has sAMB approaching +1. This is the score most relevant to our hypothesis, because visual over-reliance predicts a drift upward in sAMB when image content is added.

4. Methodology

4.1 Three-condition experimental design

Each of the 58,492 BBQ items was presented to each of two models (Qwen2-VL-2B-Instruct, Qwen2-VL-7B-Instruct) under three conditions, for a total of 6 × 58,492 = 350,952 individual inference calls. Figure 1 overviews the design.

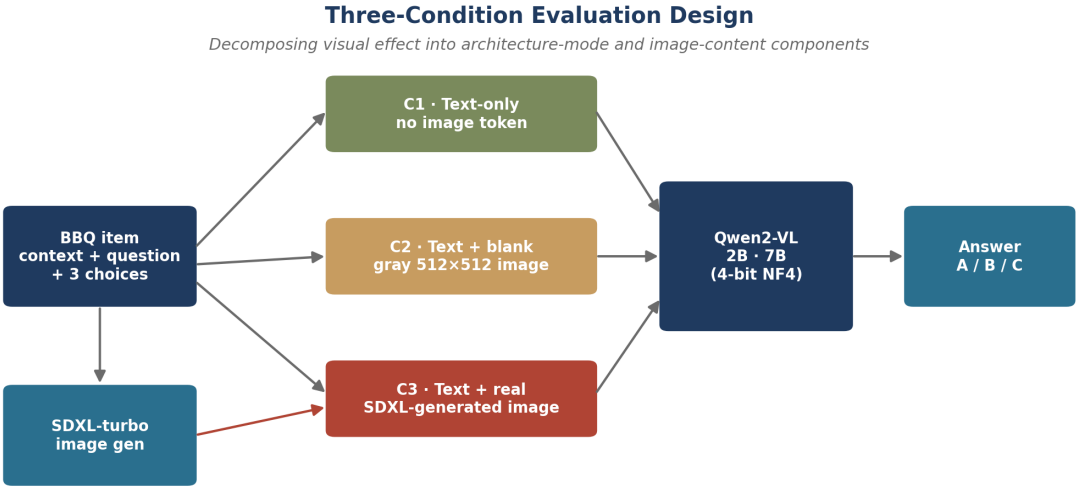

Figure 1. *Three-condition evaluation design. Each BBQ item is routed to the VLM in three parallel conditions: text-only (C1, no image token); text with a neutral blank gray 512×512 image (C2, vision encoder active, no semantic content); and text with a real SDXL-turbo-generated image of the scenario (C3). The within-item design supports paired comparisons and decomposition of the total image effect.*

4.2 Condition definitions

C1 · Text-only. The chat template contains no image placeholder. No visual tokens enter the model. This is functionally equivalent to running the underlying Qwen2 LLM without its vision components.

C2 · Text + blank image. The chat template contains an image placeholder resolved to a uniform RGB(128,128,128) gray 512×512 image. The vision encoder produces patch embeddings (near-uniform but non-zero), the merger projects them, and M-RoPE places them positionally alongside text tokens. No information about the BBQ scenario is present in the image.

C3 · Text + real image. The chat template contains an image placeholder resolved to an SDXL-turbo-generated 512×512 JPEG synthesised from the item's context. All other pipeline settings are identical to C2. The difference C2→C3 therefore isolates the effect of *semantic image content*, with vision-processing costs matched.

4.3 Image generation

Images were generated with *stabilityai/sdxl-turbo* [Podell et al. 2023] at 512×512 resolution, 2 denoising steps, guidance scale 1.2, in fp16 on an NVIDIA RTX 4090. Prompts were constructed by concatenating

the BBQ context and question, with light sanitisation (replacement of explicit age terms, removal of unsafe trigger words) and a safety suffix to steer generation toward family-appropriate output. The CLIP tokeniser truncates prompts to 77 tokens. All 58,492 images were generated in one 10.6-hour pass and reused across all 6 (model × condition) inference runs, guaranteeing that C3 differences between model runs reflect the model, not the stimulus.

Of the 58,492 attempted generations, 481 (0.82%) produced near-black or otherwise invalid outputs and were excluded from C3 analyses. The 481 rows carry an *IMAGE_MISSING* marker throughout the pipeline and are filtered from C3-involved comparisons via paired intersection.

4.4 Models and inference

Qwen2-VL-2B-Instruct (2 billion parameters) and Qwen2-VL-7B-Instruct (7 billion parameters) [Wang et al. 2024] were both loaded in 4-bit NF4 quantisation [Dettmers et al. 2023] via bitsandbytes, with compute in fp16. Greedy decoding was used (temperature 0, do_sample=False) for reproducibility. Maximum new tokens was set to 8 — sufficient for a single-letter answer. Batched inference (batch size 16) was used to amortise Python/CUDA per-call overhead.

4.5 Null hypothesis testing via paired bootstrap

The pilot used McNemar's test, which operates on the 2×2 contingency table of paired binary outcomes. Per supervisor feedback, we replace it with a paired bootstrap on the accuracy difference, which is more flexible (extends naturally to accuracy differences, bias-score differences, and per-category or per-context splits) and provides an interval estimate rather than a single p-value.

For each comparison C_a vs. C_b, let x_i and y_i be the binary correctness of the i-th item under the two conditions (i = 1 … N, using only items valid in both conditions). The observed accuracy difference is *delta* = mean(y) – mean(x). We then draw B = 10,000 bootstrap resamples of size N with replacement, compute a bootstrap *delta* for each, and report the observed *delta*, its 95% bootstrap percentile confidence interval, the bootstrap standard deviation, and a two-sided p-value (twice the proportion of bootstrap deltas whose sign opposes the observed sign). H_0 : E[y] = E[x] is rejected at the 0.05 level if the 95% CI excludes zero.

4.6 Per-category variance

In addition to overall accuracy, we report per-category mean and standard deviation across the 11 BBQ bias categories for each (model, condition) combination. Standard deviation here quantifies how uniformly the measured effect holds across bias dimensions: a large std implies the effect is category-specific, a small std implies it is general.

5. Results

5.1 Overall accuracy and bias scores

Model	Condition	Accuracy	sAMB	sDIS
Qwen2-VL-2B	C1 · text only	0.5463	+0.079	+0.014
	C2 · + blank image	0.5239	+0.076	+0.015
	C3 · + real image	0.4981	+0.084	+0.027
Qwen2-VL-7B	C1 · text only	0.8044	+0.098	–0.003
	C2 · + blank image	0.6981	+0.110	0.000

	C3 · + real image	0.6563	+0.113	+0.007

Table 1. *Overall accuracy and BBQ bias scores per model and condition. All values computed over all 58,492 items (58,011 for C3, after removing IMAGE_MISSING rows). Positive sAMB indicates stereotype-consistent bias in ambiguous contexts.*

Accuracy drops monotonically from C1 to C2 to C3 for both models. Qwen2-VL-2B loses –4.83 pp overall (54.63% → 49.81%), and Qwen2-VL-7B loses –14.81 pp (80.44% → 65.63%). The larger model suffers the larger drop despite starting from a higher baseline — a counterintuitive pattern we discuss in §6.1.

The sAMB score rises monotonically C1 → C2 → C3 for Qwen2-VL-7B (+0.098 → +0.110 → +0.113), meaning that each step toward content-richer input increases the tendency to commit to stereotyped answers in ambiguous conditions. sDIS remains near zero across conditions, confirming that when the text provides sufficient evidence, both models rarely choose stereotyped over counter-stereotyped responses — as prior BBQ work has consistently reported.

5.2 Decomposing the visual effect

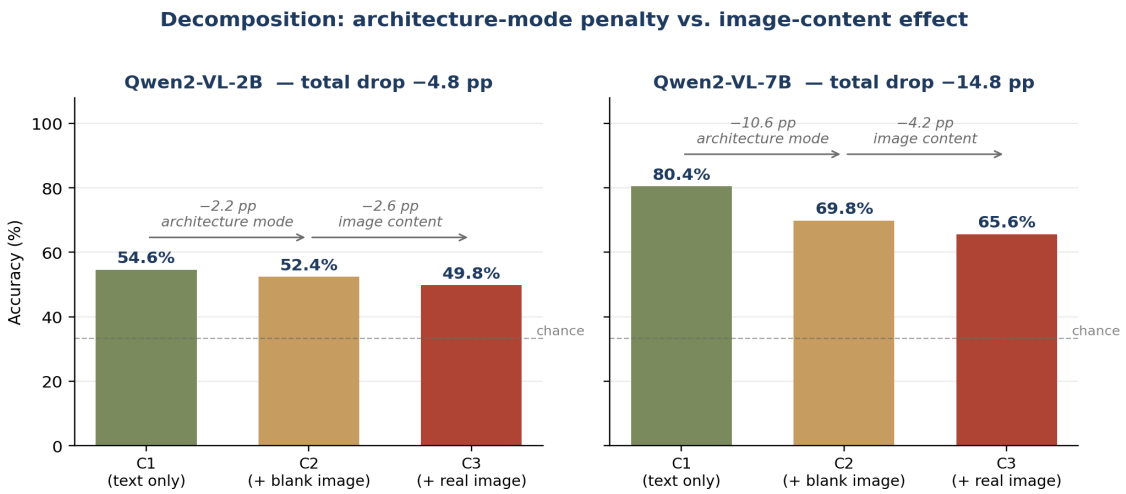

Figure 2. *Decomposition of the total C1→C3 accuracy drop into an architecture-mode penalty (C1→C2, left-hand arrow on each panel) and an image-content effect (C2→C3, right-hand arrow). For Qwen2-VL-7B, the architecture-mode component (10.6 pp) is substantially larger than the content component (4.2 pp).*

Figure 2 makes the decomposition visible. On the 7B model, the total 14.8 pp drop splits into 10.6 pp (72% of total) attributable to C1→C2 and 4.2 pp (28%) attributable to C2→C3. On the 2B model, the split is 2.2 pp vs. 2.6 pp, roughly balanced. **For the capability-paradox finding originally reported in the pilot, most of the 7B-over-2B excess is architecture-mode, not content-driven**: the 7B model's vision encoder is disproportionately disruptive to its otherwise strong text reasoning.

5.3 Paired bootstrap tests

Table 2 reports the paired bootstrap results for the three key comparisons on both models, with separate splits for ambiguous and disambiguating contexts.

Model	Comparison	Context	Diff. acc.	95% CI	p
Qwen2-VL-2B	C1 vs C3 (total image effect)	all	–0.048	[–0.051, –0.045]	<0.0001
		ambig	–0.141	[–0.145, –0.136]	<0.0001
		disambig	+0.044	[+0.040, +0.048]	<0.0001
	C2 vs C3 (image-content effect)	all	–0.026	[–0.029, –0.023]	<0.0001

		ambig	–0.086	[–0.090, –0.082]	<0.0001
	C1 vs C2 (architecture mode)	all	–0.022	[–0.024, –0.021]	<0.0001
		ambig	–0.055	[–0.057, –0.052]	<0.0001
Qwen2-VL-7B	C1 vs C3 (total image effect)	all	–0.150	[–0.154, –0.147]	<0.0001
		ambig	–0.303	[–0.308, –0.297]	<0.0001
		disambig	+0.002	[–0.000, +0.004]	0.083
	C2 vs C3 (image-content effect)	all	–0.044	[–0.046, –0.041]	<0.0001
		ambig	–0.077	[–0.082, –0.072]	<0.0001
	C1 vs C2 (architecture mode)	all	–0.106	[–0.109, –0.103]	<0.0001
		ambig	–0.225	[–0.230, –0.220]	<0.0001

Table 2. *Paired bootstrap tests on accuracy differences. 10,000 resamples per comparison; 95% percentile CI. Only rows valid in both conditions of a comparison are included (paired intersection). All but one comparison reject H_0 at the 0.05 level; the exception is the 7B C1-vs-C3 disambiguating split (p = 0.083), which reinforces the ambiguity-specific nature of the image-content effect.*

Two properties of Table 2 are diagnostic. First, **every ambiguous-context comparison rejects H_0 with a negative difference**: image presence systematically reduces accuracy on items where the correct answer is "unknown". Second, the **7B C1-vs-C3 disambiguating comparison is the single test in the whole table that fails to reject H_0** (diff = +0.002, p = 0.083). When text provides sufficient evidence, the real SDXL image is not reliably harmful. This is strong evidence that the content-driven effect is ambiguity-specific and not a general multimodal processing degradation — consistent with the prediction in §2.3 that visual bias emerges only when textual evidence is insufficient.

5.4 Per-category sAMB drift

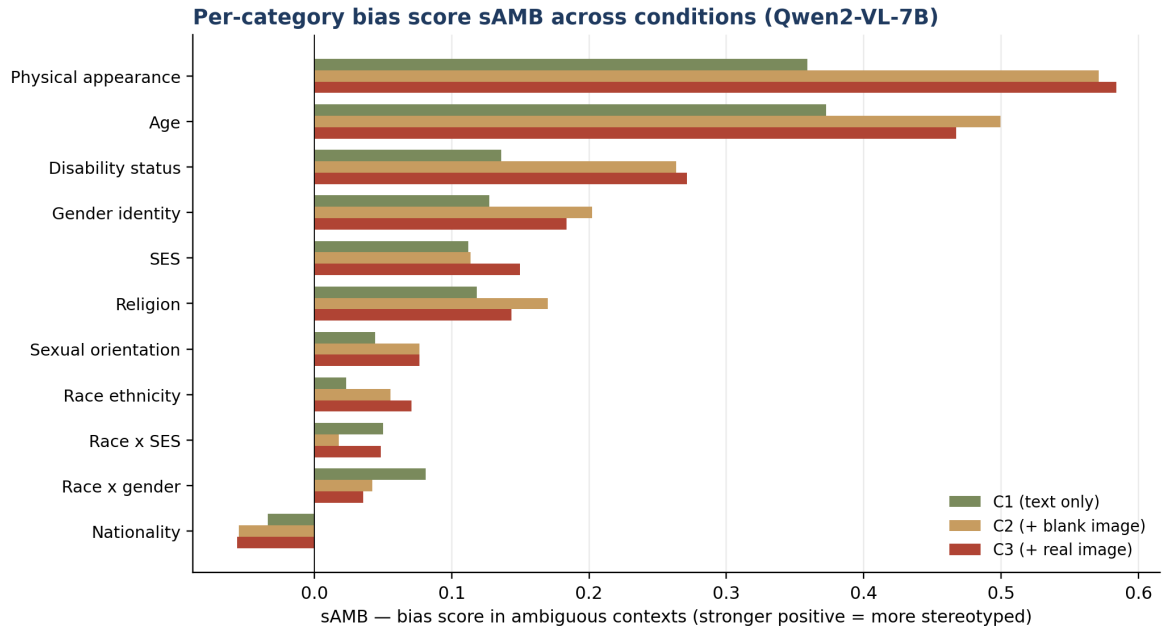

Figure 3. *Per-category sAMB for Qwen2-VL-7B across the three conditions. Categories are sorted by their C3 sAMB score. The categories with the largest C1→C3 drift (Physical appearance, Age, Disability status, Gender identity) are those whose stereotypes are most visually inferable; categories driven by non-visual attributes (Nationality, Race × gender) show near-zero drift.*

Figure 3 and Table 3 support a content-specific interpretation of the C2→C3 residual. Physical appearance moves from sAMB = 0.359 (C1) to 0.584 (C3), an absolute increase of 0.225 — far larger than any non-visual category. Age shows a similarly large jump (+0.094). By contrast, Nationality, a category where ambient stereotypes are linguistic-cultural rather than visual, shows a *decrease* in sAMB from –0.034 (C1) to –0.056 (C3), an essentially null effect. This category-level pattern is what a content-driven stereotype mechanism would predict, and is inconsistent with a purely architecture-mode account of the C2→C3 residual.

5.5 Accuracy by context condition and per-category variance

Model	Condition	Ambig (mean ± std)	Disambig (mean ± std)	n_cat
Qwen2-VL-2B	C1	0.337 ± 0.077	0.752 ± 0.057	11
	C2	0.285 ± 0.071	0.761 ± 0.061	11
	C3	0.192 ± 0.074	0.798 ± 0.055	11
Qwen2-VL-7B	C1	0.674 ± 0.086	0.926 ± 0.053	11
	C2	0.451 ± 0.098	0.943 ± 0.046	11
	C3	0.384 ± 0.080	0.931 ± 0.051	11

Table 3. *Mean ± standard deviation of per-category accuracy across the 11 BBQ bias categories, by context condition. Ambiguous-context accuracy drops sharply C1→C3 on both models; disambiguating-context accuracy is stable.*

The ambiguous-disambiguating split in Table 3 mirrors the bootstrap findings of Table 2 but at the category-aggregated level. Disambiguating-context accuracy is stable across all three conditions on both models (0.752 → 0.761 → 0.798 for the 2B model; 0.926 → 0.943 → 0.931 for the 7B model). The entire accuracy drop is borne by ambiguous-context items. The per-category standard deviation in the ambiguous condition is 0.07–0.10, indicating that the effect is broadly distributed rather than driven by one outlier category.

6. Discussion

6.1 The capability paradox, revisited

The pilot study reported that the 7B model suffered a larger visual drop than the 2B model despite its higher baseline — a “capability paradox” suggesting that stronger visual grounding comes at the cost of susceptibility to visual bias. The decomposition in Figure 2 reframes this finding. The 7B model's excess drop is driven chiefly by its architecture-mode penalty (10.6 pp, vs. 2.2 pp on the 2B model), not by its image-content sensitivity (4.2 pp vs. 2.6 pp). In other words: the 7B model’s problem is not that it “trusts the image too much” — it is that simply running its vision branch with *any* input degrades its text reasoning, and that degradation scales with model size.

A plausible mechanism is that the M-RoPE positional recomputation and the insertion of ~250 near-uniform visual tokens into the attention stream dilute the attention paid to text tokens, and that this dilution costs more in models with stronger text reasoning. Alternatively, the 7B model’s larger LLM backbone may be more sensitive to mild out-of-distribution shifts in its input sequence statistics. Distinguishing these hypotheses would require ablations beyond the scope of the present study.

6.2 Ambiguity as the trigger for visual bias

The near-null C3 effect in disambiguating contexts (7B diff = +0.002, p = 0.083) is, if anything, our cleanest positive result, because it clearly separates the conditions under which the model relies on visual information from those under which it does not. One natural framing is Bayesian: the model combines a text-derived likelihood with an image-derived prior. When the text likelihood is sharp (disambiguating), the posterior is dominated by text and image effects are negligible; when the text likelihood is flat (ambiguous), the posterior is dominated by the visual prior, and if that prior encodes stereotypes, the model's output inherits them. This matches the core pattern reported by the original BBQ paper — bias manifests chiefly when the model lacks sufficient evidence to be right.

6.3 Why the per-category pattern matters

Figure 3 rules out a purely architecture-mode explanation for the residual C2→C3 effect. If the residual drop were simply additional architecture-mode noise from more visually complex patches, it would have no reason to concentrate in visually inferable categories. That it concentrates specifically in Physical appearance, Age, Disability status, and Gender identity — categories where diffusion-generated depictions carry clear demographic signals — is direct evidence for a stereotype-driven mechanism acting on semantic image content. For this reason, the residual C2→C3 effect is the component of the total drop most reasonably called "visual over-reliance" in the stereotype-specific sense; the C1→C2 component should be given a different name.

6.4 Limitations

Three caveats deserve explicit mention. First, 4-bit NF4 quantisation may amplify noise in sensitive comparisons; full-precision inference (not feasible on the 24 GB RTX 4090 used here, particularly for 7B) could yield slightly different effect sizes. Second, 0.82% of SDXL generations produced invalid outputs and were excluded from C3; this is documented per condition in the data-coverage section of the accompanying *summary_report.txt*, and the paired-intersection design means those 481 rows are excluded from all C3-involved comparisons in a principled way. Third, only two model sizes from the Qwen2-VL family are tested; the architecture-mode vs. content-effect decomposition should be verified on other LVLM families (LLaVA, PaliGemma, Gemma 3, InternVL) before treating the decomposition ratio as general.

7. Conclusion

The pilot study replicated at full scale here found a large accuracy drop when BBQ items were paired with AI-generated images. The three-condition design reported in this paper shows that this drop conflates two distinct phenomena: an architecture-mode penalty, incurred simply by activating the vision encoder with any content, and an image-content effect, incurred specifically by semantic stereotypes in the image. For Qwen2-VL-7B, the architecture-mode penalty (–10.6 pp) accounts for roughly 72% of the total drop; the content effect (–4.2 pp) accounts for the remaining 28%. Only the content effect is category-specific, ambiguity-triggered, and aligned with the BBQ bias score sAMB — and only it should be interpreted as stereotype-driven visual bias in the narrow sense.

The practical implication for deployment is that two independent mitigations are needed. An architecture-mode penalty suggests prompting strategies that make the vision-encoder path optional when the task is primarily textual (e.g. conditional image routing based on task classification). An image-content effect suggests stereotype-specific debiasing (along lines explored by Jung et al. 2024 and Pang et al. 2025) targeted at ambiguous-context inputs. Conflating the two under a single "visual bias" label obscures which intervention is relevant.

Full code, generated images, prediction CSVs, and analysis outputs are available at: https://github.com/logvisionn/bbq-visual-overreliance.

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